Unit 5 Progress Check: MCQ

$2\,X + Y_2 \rightarrow X_2Y_2$

A chemist is studying the reaction between the gaseous chemical species X and Y_2, represented by the equation above. Initial rates of reaction are measured at various concentrations of reactants. The results are recorded in the following table.

Experiment	$[X]_i$	$[Y_2]_i$	Initial Rate of Appearance of X_2Y_2 (M/s)
1	0.15	0.10	32
2	0.15	0.20	64
3	0.30	0.20	128

1. Given the information in the table above, which of the following is the experimental rate law?

- (A) Rate $= k[X][Y_2]$ ✓
- (B) Rate $= k\left[X\right]^2\left[Y_2\right]$
- (C) Rate $= k[X][Y_2]^2$
- (D) Rate $= k[X]^2[Y_2]^2$

2. Based on the information above, determine the initial rate of disappearance of X in experiment 1.

- (A) 16 M/s
- (B) 32 M/s
- (C) 64 M/s ✓
- (D) 128 M/s

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3. A second chemist repeated the three experiments and observed that the reaction rates were considerably greater than those measured by the first chemist even though the concentrations of the reactants and the temperature in the laboratory were the same as they were for the first chemist. Which of the following is the best pairing of a claim about a most likely cause for the greater rates measured by the second chemist and a valid justification for that claim?

(A) The pressures of the gases used by the second chemist must have been lower than those used by the first chemist, thus the collisions between reacting particles were less frequent than they were in the first chemist's experiments.

(B) The pressures of the gases used by the second chemist must have been lower than those used by the first chemist, thus the number of collisions with sufficient energy to cause reaction was lower than it was in the first chemist's experiments.

(C) The second chemist must have added a catalyst for the reaction, thus providing a different reaction pathway for the reactant particles to react with an activation energy that was lower than that of the uncatalyzed reaction in the first chemist's experiments. ✓

(D) The second chemist must have added a catalyst for the reaction, thus providing energy to reactant particles to increase their rate of reaction compared to their rate of reaction in the first chemist's experiments.

4. $C_{12}H_{22}O_{11}(aq) + H_2O(l) \rightarrow 2\ C_6H_{12}O_6(aq)$

The chemical equation shown above represents the hydrolysis of sucrose. Under certain conditions, the rate is directly proportional to the concentration of sucrose. Which statement supports how a change in conditions can increase the rate of this reaction?

(A) Increasing the amount of water in which the sugar is dissolved will increase the frequency of collisions between the sucrose molecules and the water molecules resulting in an increase in the rate of hydrolysis.

(B) Decreasing the temperature will increase the frequency of the collisions between the sucrose molecules and the water molecules resulting in an increase in the rate of hydrolysis.

(C) Increasing the concentration of sucrose will increase the rate of hydrolysis by increasing the frequency of the collisions between the sucrose and the water molecules. ✓

(D) Decreasing the concentration of sucrose will increase the rate of hydrolysis by increasing the frequency of the collisions between the sucrose and the water molecules.

Unit 5 Progress Check: MCQ

5. $2\,NO(g) + 2\,H_2(g) \rightarrow N_2(g) + 2\,H_2O(g)$

Trial	Initial Concentration of NO (M)	Initial Concentration of H_2 (M)	Initial Rate of Formation of N_2 (M/s)
1	0.10	0.10	1.2×10^{-3}
2	0.10	0.20	2.4×10^{-3}

The information in the data table above represents two different trials for an experiment to study the rate of the reaction between $NO(g)$ and $H_2(g)$, as represented by the balanced equation above the table. Which of the following statements provides the correct explanation for why the initial rate of formation of N_2 is greater in trial 2 than in trial 1 ? Assume that each trial is carried out at the same constant temperature.

(A) The activation energy of the reaction is smaller in trial 2 than it is in trial 1.

(B) The frequency of collisions between reactant molecules is greater in trial 2 than it is in trial 1. ✓

(C) The value of the rate constant for the reaction is smaller in trial 2 than it is in trial 1.

(D) The value of the rate constant for the reaction is greater in trial 2 than it is in trial 1.

6. $Zn(s) + 2\,HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$

$Zn(s)$ reacts with $HCl(aq)$ according to the equation shown above. In trial 1 of a kinetics experiment, a 5.0 g piece of $Zn(s)$ is added to 100 mL of $0.10\,M$ $HCl(aq)$. The rate of reaction between $Zn(s)$ and $HCl(aq)$ is determined by measuring the volume of $H_2(g)$ produced over time. In trial 2 of the experiment, 5.0 g of powdered $Zn(s)$ is added to 100 mL of $0.10\,M$ $HCl(aq)$. Which trial will have a faster initial rate of reaction and why?

(A) Trial 1, because there is a higher concentration of $Zn(s)$ in the reaction mixture.

(B) Trial 1, because the sample of $Zn(s)$ has less surface area for the reaction to take place.

(C) Trial 2, because there is a higher concentration of $HCl(aq)$ in the reaction mixture.

(D) Trial 2, because the sample of $Zn(s)$ has a greater surface area for the reaction to take place. ✓

Unit 5 Progress Check: MCQ

7. $CH_3I + NaOH \rightarrow CH_3OH + NaI$

The rate of the reaction represented by the chemical equation shown above is expressed as $rate = k[CH_3I][NaOH]$. Based on this information, which of the following claims is correct?

(A) The reaction will proceed at a slower rate with increasing temperature.

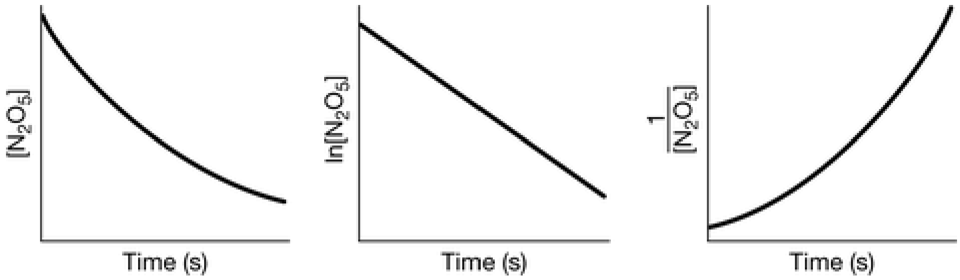
(B) The rate of the reaction will double when the concentrations of both CH_3I and $NaOH$ are doubled.

(C) The rate of the reaction will double if the concentration of CH_3I is doubled while keeping the concentration of $NaOH$ constant. ✓

(D) A larger amount of CH_3OH will be produced if the concentrations of CH_3I and $NaOH$ are halved.

8. $2\,N_2O_5\,(g) \rightarrow 4\,NO_2\,(g) + O_2\,(g)$

For the reaction represented by the equation above, the concentration of N_2O_5 was measured over time. The following graphs were created using the data.



Based on the graphs above, what is the order of the reaction with respect to N_2O_5 ?

(A) Zeroth order

(B) First order ✓

(C) Second order

(D) Third order

Unit 5 Progress Check: MCQ

9. $^{214}_{83}\mathrm{Bi} \rightarrow \, ^{214}_{84}\mathrm{Po} + \, ^{0}_{-1}e$

Bismuth-214 undergoes first-order radioactive decay to polonium-214 by the release of a beta particle, as represented by the nuclear equation above.

Which of the following quantities plotted versus time will produce a straight line?

(A) [Bi]

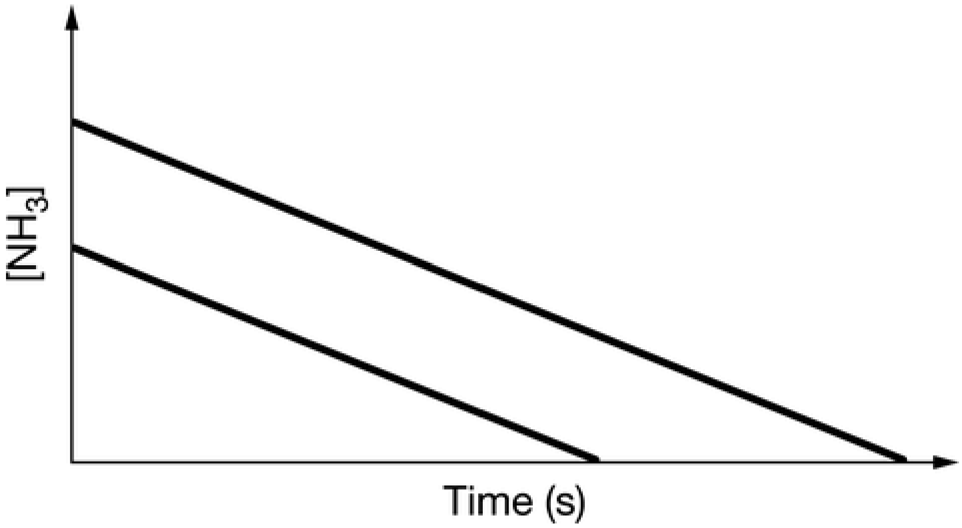
(B) [Po]

(C) ln[Bi] ✓

(D) $\frac{1}{[\mathrm{Bi}]}$

10. $2\,\mathrm{NH_3}(g) \xrightarrow{\text{catalyst}} \mathrm{N_2}(g) + 3\,\mathrm{H_2}(g)$

The catalyzed decomposition of $\mathrm{NH_3}(g)$ at high temperature is represented by the equation above. The rate of disappearance of $\mathrm{NH_3}(g)$ was measured over time for two different initial concentrations of $\mathrm{NH_3}(g)$ at a constant temperature. The data are plotted in the graph below.



On the basis of the data in the graph, which of the following best represents the rate law for the catalyzed decomposition of $\mathrm{NH_3}(g)$?

(A) Rate $= k$ ✓

(B) Rate $= k[NH_3]$

(C) Rate $= k\left[NH_3\right]^2$

(D) Rate $= k[N_2][H_2]$

11.

Step 1:	$H_2O_2 + I^- \rightarrow IO^- + H_2O$
Step 2:	$H_2O_2 + IO^- \rightarrow H_2O + O_2 + I^-$

The mechanism for a chemical reaction is shown above. Which of the following statements about the overall reaction and rate laws of the elementary reactions is correct?

(A) The chemical equation for the overall reaction is $2H_2O_2 + I^- \rightarrow 2H_2O + O_2 + I^-$, and the rate law for elementary step 2 is $rate = k[H_2O_2][IO^-]$.

(B) The chemical equation for the overall reaction is $H_2O_2 + IO^- \rightarrow H_2O + O_2 + I^-$, and the rate law for elementary step 2 is $rate = k[H_2O_2]^2[IO^-]$.

(C) The chemical equation for the overall reaction is $2H_2O_2 \rightarrow 2H_2O + O_2$, and the rate law for elementary step 1 is $rate = k[H_2O_2][I^-]$. ✓

(D) The chemical equation for the overall reaction is $2H_2O_2 \rightarrow 2H_2O + O_2$, and the rate law for elementary step 1 is $rate = k[H_2O_2]^2$.

12.

Step 1:	$N_2O_5 \rightarrow NO_2 + NO_3$	(*slow*)
Step 2:	$NO_2 + NO_3 \rightarrow NO_2 + NO + O_2$	(*fast*)
Step 3:	$NO + N_2O_5 \rightarrow 3NO_2$	(*fast*)

A proposed reaction mechanism for the decomposition of $N_2O_5(g)$ is shown above. Based on the proposed mechanism, which of the following correctly identifies both the chemical equation and the rate law for the overall reaction?

A The chemical equation for the overall reaction is $NO(g) + N_2O_5(g) \rightarrow 3\,NO_2(g)$, and the rate law is $rate = k[NO][N_2O_5]$.

B The chemical equation for the overall reaction is $2\,N_2O_5(g) \rightarrow 4\,NO_2(g) + O_2(g)$, and the rate law is $rate = k[N_2O_5]$. ✓

C The chemical equation for the overall reaction is $2\,N_2O_5(g) \rightarrow 4\,NO_2(g) + O_2(g)$, and the rate law is $rate = k[N_2O_5]^2$.

D The chemical equation for the overall reaction is $N_2O_5(g) + NO(g) + NO_3(g) \rightarrow 4\,NO_2(g) + O_2(g)$, and the rate law is $rate = k[N_2O_5][NO][NO_3]$.

13.

Step 1:	?	(*slow*)
Step 2:	$NO_2(g) + F(g) \rightarrow NO_2F(g)$	(*fast*)
Overall:	$2\,NO_2(g) + F_2(g) \rightarrow 2\,NO_2F(g)$	

The overall reaction represented above is proposed to take place through two elementary steps. Which of the following statements about the chemical equation for step 1 and the rate law for the overall reaction is correct?

A The chemical equation for step 1 is $NO_2(g) + F_2(g) \rightarrow NO_2F(g) + F(g)$, and the rate law for the overall reaction is $rate = k[NO_2][F_2]$. ✓

B The chemical equation for step 1 is $NO_2(g) + F(g) \rightarrow NO_2F(g) + F_2(g)$, and the rate law for the overall reaction is $rate = k[NO_2][F]$.

C The chemical equation for step 1 is $2\,NO_2(g) + F_2(g) \rightarrow 2\,NO_2F(g)$, and the rate law for the overall reaction is $rate = k[NO_2][F]$.

D The chemical equation for step 1 is $3\,NO_2(g) + F_2(g) + F(g) \rightarrow 3\,NO_2F(g)$, and the rate law for the overall reaction is $rate = k[NO_2]^2[F_2]$.

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14.

Reaction A:	$O + O \rightarrow O_2$
Reaction B:	$C_2H_4 + C_2H_4 \rightarrow C_4H_8$
Reaction C:	$CO + O_2 \rightarrow CO_2 + O$
Reaction D:	$CH_3I + Br^- \rightarrow CH_3Br + I^-$

The equations shown above represent four elementary reactions. Which of the following identifies the reaction in which the number of successful collisions and reaction rate are independent of the orientation of the reactants and explains why?

(A) Reaction A, because the electron clouds of the O atoms are distributed symmetrically. ✓

(B) Reaction B, because each C_2H_4 molecule has a double bond.

(C) Reaction C, because both CO and O_2 are linear molecules.

(D) Reaction D, because the reactant Br^- and the product I^- are negatively charged.

15.

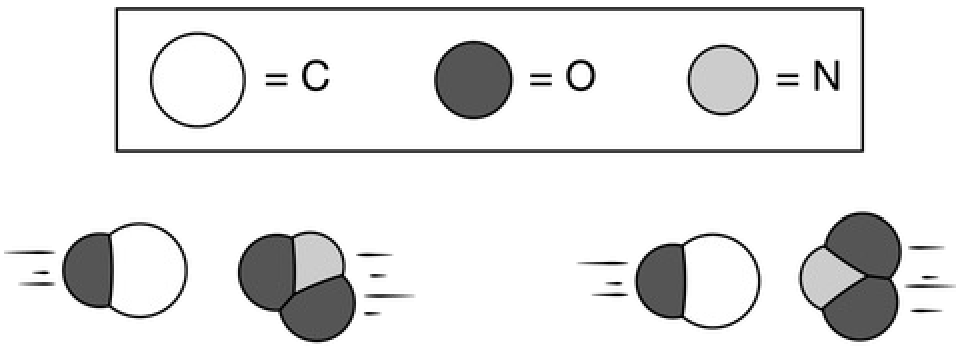

Diagram 1 Diagram 2

The two diagrams above represent collisions that take place at the same temperature between a CO molecule and an NO_2 molecule. The products are CO_2 and NO. Which diagram most likely represents an effective collision, and why?

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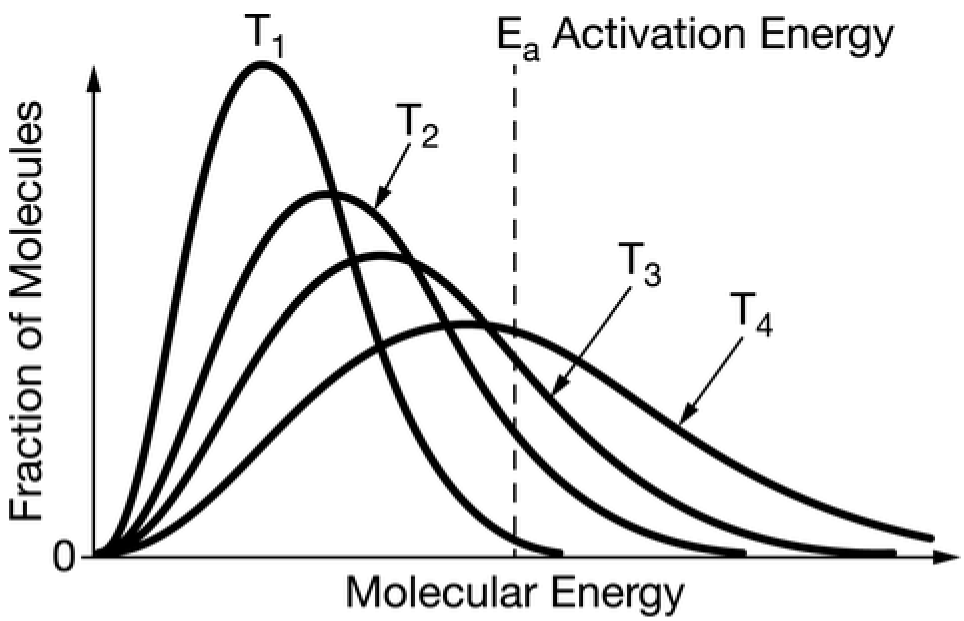
(A) Diagram 1 represents an effective collision because the molecules have about the same size and the same energy, which leads to a larger rate constant (***k***).

(B) Diagram 1 represents an effective collision because the two molecules have the proper orientation to form a new $C - O$ bond as long as they possess enough energy to overcome the activation energy barrier. ✓

(C) Diagram 2 represents an effective collision because the molecules are aligned head-to-head and that maximizes the overlap between their atomic orbitals, forming more products.

(D) Diagram 2 represents an effective collision because the oxygen atoms are the most electronegative and will minimize their repulsions when oriented away from each other.

16.



The diagram above shows the distribution of molecular energies for equimolar samples of a reactant at different temperatures. Based on the diagram, at which temperature will the reactant be consumed at the fastest rate, and why?

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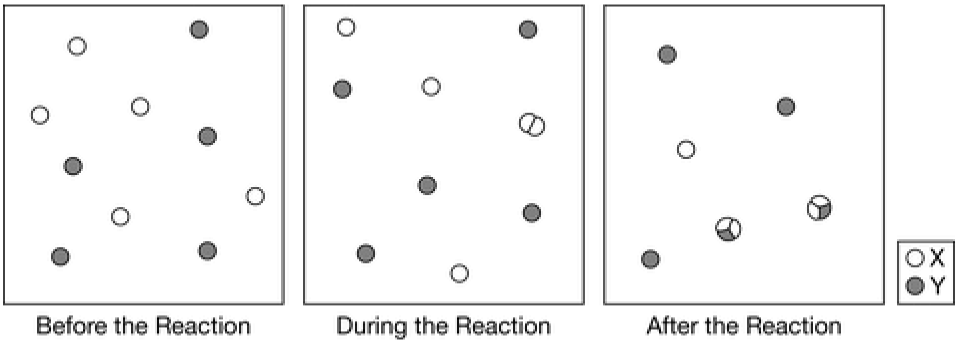
(A) At T_1, because a larger fraction of the molecules have about the same energy.

(B) At T_2, because at this temperature most of the molecules undergo collisions frequently.

(C) At T_3, because at this temperature the rate of consumption for about half of the molecules is determined by their orientation.

(D) At T_4, because a larger fraction of the molecules have an energy that is equal to or greater than the activation energy. ✓

17. 

Which of the following best describes the elementary step(s) in the reaction mechanism represented in the diagram above?

(A)

One step:	$2\,X(g) + Y(g) \rightarrow X_2Y(g)$

(B)

One step:	$X_2(g) + Y(g) \rightarrow X_2Y(g)$

(C)

Two steps:	Step 1:	$X(g) + Y(g) \rightarrow XY(g)$
	Step 2:	$X(g) + XY(g) \rightarrow X_2Y(g)$

(D) ✓

Two steps:	Step 1:	$2\,X(g) \rightarrow X_2(g)$
	Step 2:	$X_2(g) + Y(g) \rightarrow X_2Y(g)$

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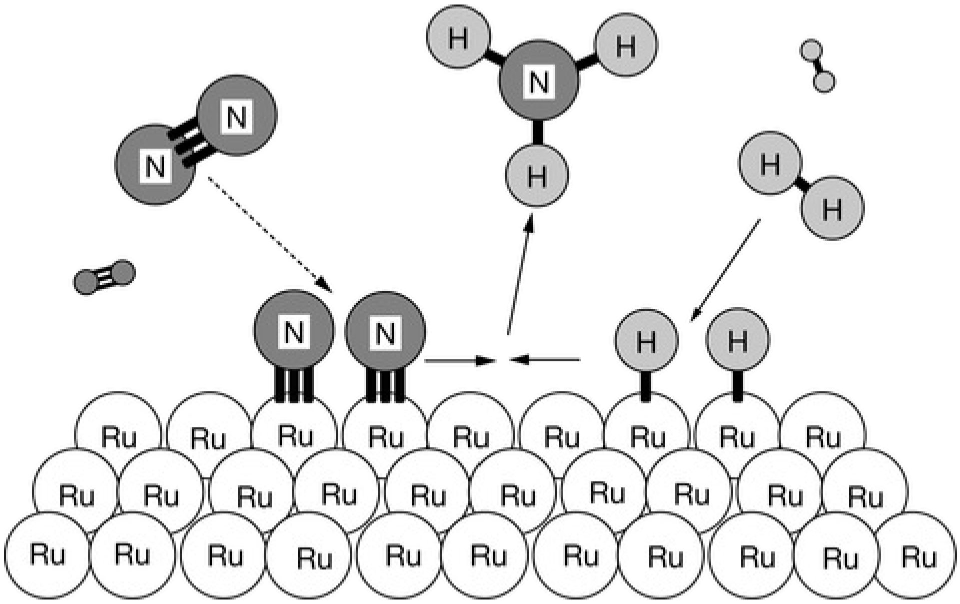
18.



The diagram above shows the progress of the chemical reaction for the synthesis of ammonia from its elements. The adsorption of the N_2 molecules on the surface of Ru weakens the triple bond between the two N atoms. Based on the diagram, what is the role of Ru in this process?

(A) Ru is a catalyst. ✓

(B) Ru is a reactant.

(C) Ru is a product.

(D) Ru is an intermediate.

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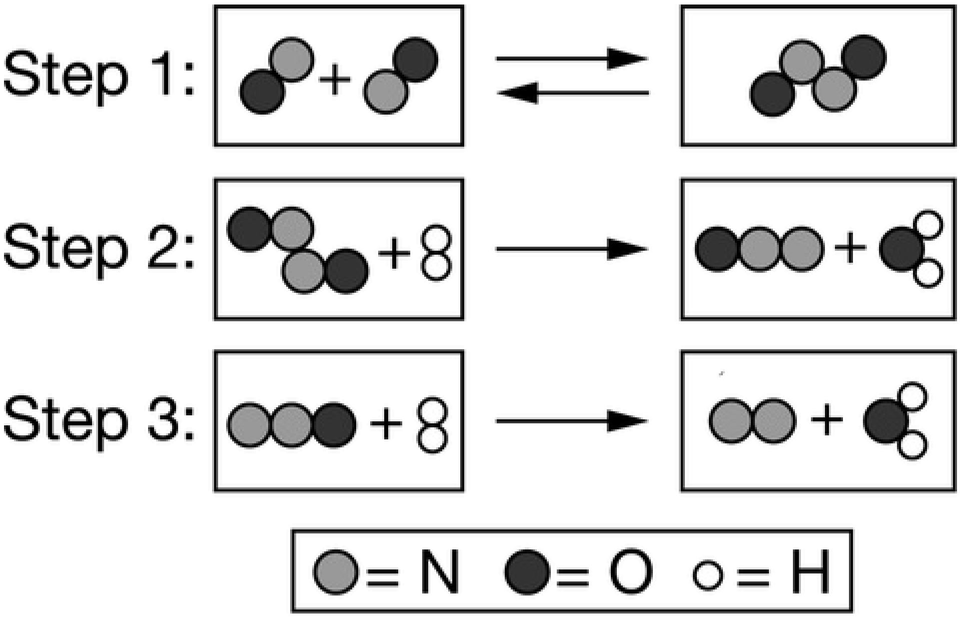
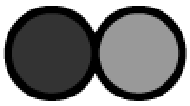
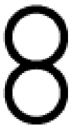
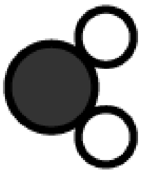
19.

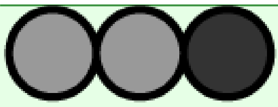


The proposed mechanism for a reaction involves the three elementary steps represented by the particle models shown above. On the basis of this information, which of the following models represents an intermediate in the overall reaction?

(A) 

(B) 

(C) 

(D)  ✓

20. $O_3(g) + O(g) \rightarrow 2\ O_2(g)$

The decomposition of O_3 occurs according to the balanced equation above. In the presence of NO, the decomposition proceeds in two elementary steps, as represented by the following mechanism.

Step 1:	$O_3 + NO \rightarrow NO_2 + O_2$	(slow)
Step 2:	$NO_2 + O \rightarrow NO + O_2$	(fast)

Based on the information, which of the following is the rate law?

Unit 5 Progress Check: MCQ

(A) Rate $= k[O_3][NO]$ ✓

(B) Rate $= k[NO_2][O_2]$

(C) Rate $= k[NO_2][O]$

(D) Rate $= k[NO][O]$

21.

Step 1:	$2\ NO_2\,(g) \rightarrow NO_3\,(g) + NO\,(g)$	(*slow*)
Step 2:	$NO_3(g) + CO(g) \rightarrow NO_2(g) + CO_2(g)$	(*fast*)

A proposed two-step mechanism for the chemical reaction $NO_2(g) + CO(g) \rightarrow NO(g) + CO_2(g)$ is shown above. Which of the following equations is a correct rate law that is consistent with the elementary steps in the mechanism?

(A) Rate $= k[NO_2]$

(B) Rate $= k[NO_2]^2$ ✓

(C) Rate $= k[NO_3][CO]$

(D) Rate $= k[NO_2][CO]$

22.

Step 1:	$H_2 + IBr \rightarrow HI + HBr$	(slow)
Step 2:	$HI + IBr \rightarrow I_2 + HBr$	(fast)

A proposed mechanism for the reaction $H_2 + 2\,IBr \rightarrow I_2 + 2\,HBr$ is shown above. Two experiments were performed at the same temperature but with different initial concentrations. Based on this information, which of the following statements is correct?

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(A) The rate of the reaction will undergo a 4-fold increase in the experiment in which the initial concentrations of both HI and IBr were doubled.

(B) The rate of the reaction will undergo a 2-fold increase in the experiment in which the initial concentrations of both HI and IBr were doubled.

(C) The rate of the reaction will undergo a 4-fold increase in the experiment in which the initial concentrations of both H_2 and IBr were doubled. ✓

(D) The rate of the reaction will undergo a 8-fold increase in the experiment in which the initial concentrations of both H_2 and IBr were doubled.

23.

Step 1:	$2\,NO \rightleftarrows (NO)_2$	(fast)
Step 2:	$(NO)_2 + O_2 \rightleftarrows 2\,NO_2$	(slow)

The elementary steps in a proposed mechanism for the reaction $2\,NO(g) + O_2(g) \rightarrow 2\,NO_2(g)$ are represented by the equations above. Which of the following is the rate law for the overall reaction that is consistent with the proposed mechanism?

(A) rate $= k[NO]^2$

(B) rate $= k[NO][O_2]$

(C) rate $= k[NO]^2[O_2]$ ✓

(D) rate $= k[(NO)_2][O_2]$

24. $H_2(g) + I_2(g) \rightarrow 2\,HI(g)$

For the reaction between H_2 and I_2, shown above, the following two-step reaction mechanism is proposed.

Step 1:	$I_2 \rightleftarrows 2\,I$	(fast equilibrium)
Step 2:	$H_2 + 2\,I \rightarrow 2\,HI$	(slow)

What is the rate law expression for this reaction if the second step is rate determining?

(A) Rate $= k[I_2]$

(B) Rate $= k\left[H_2\right][I]^2$

(C) Rate $= k[H_2][I_2]$ ✓

(D) Rate $= k\left[I_2\right][I]^2$

25.

Step 1:	$2\,X(g) \rightleftarrows X_2(g)$	(fast)
Step 2:	$X_2(g) + Y(g) \rightarrow X_2Y(g)$	(slow)

The rate law for the hypothetical reaction $2\,X\,(g) + Y(g) \rightarrow X_2Y\,(g)$ is consistent with the mechanism shown above. Which of the following mathematical equations provides a rate law that is consistent with this mechanism?

(A) rate $= k[X]^2$

(B) rate $= k\,[X_2]\,[Y]$

(C) rate $= k[X]^2\,[Y]$ ✓

(D) rate $= k\,[2X]\,[Y]$

26.

Step 1:	$HCOOH + H_2SO_4 \rightarrow HCOOH_2{}^+ + HSO_4{}^-$
Step 2:	$HCOOH_2{}^+ \rightarrow HCO^+ + H_2O$
Step 3:	$HCO^+ + HSO_4{}^- \rightarrow CO + H_2SO_4$

The elementary steps in a proposed mechanism for the decomposition of $HCOOH$ are represented above. Which of the following identifies the catalyst in the overall reaction and correctly justifies the choice?

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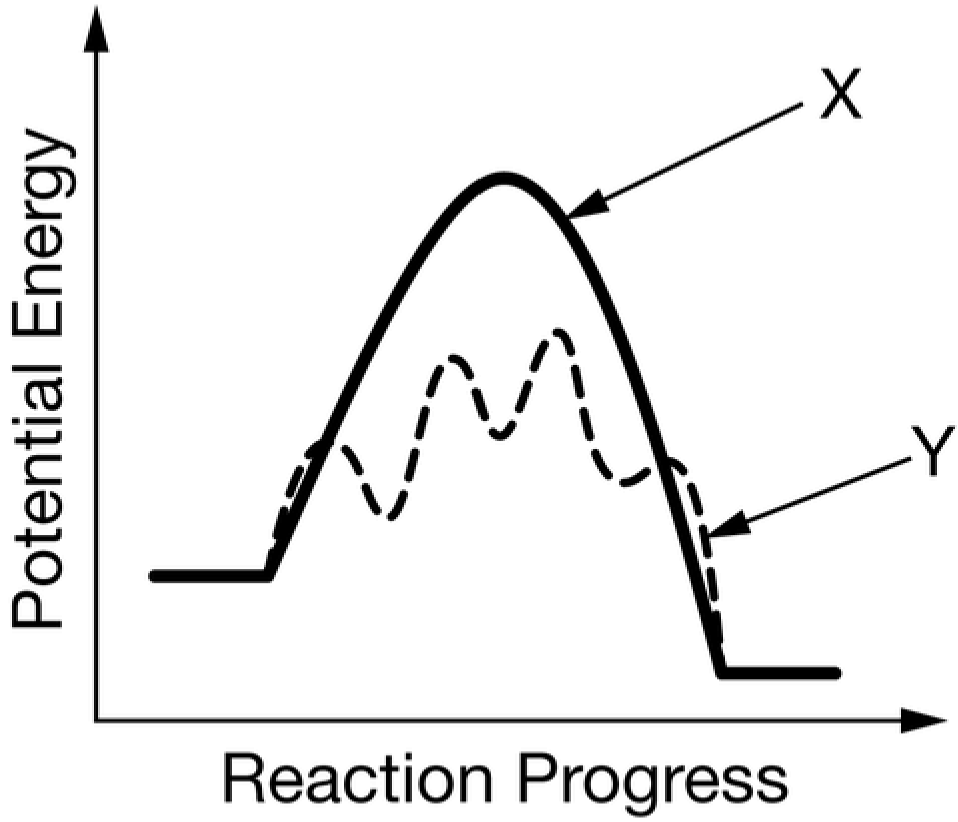
(A) H_2SO_4, because it is consumed in the first step of the mechanism and regenerated in a later step. ✓

(B) HCOOH, because increasing the concentration of HCOOH increases the reaction rate.

(C) $HCOOH_2{}^+$, because it is produced in the first step of the mechanism and consumed in a later step.

(D) CO, because it is produced in the final step of the reaction mechanism.

27.



The diagram above shows the reaction energy profiles for a reaction with and without a catalyst. Which of the following identifies the reaction energy profile for the catalyzed reaction, and why?

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(A) Profile X, because a catalyst minimizes the number of elementary steps required for a reaction to proceed.

(B) Profile X, because the activation energy for the reverse reaction is greater than for the forward reaction, which increases its rate.

(C) Profile Y, because an increase in the number of transition states increases the rate of the reaction.

(D) Profile Y, because it introduces a different reaction path that reduces the activation energy. ✓